

USA 25 Barnett Lost In Translation Exploiting Unicode compressed (English translation)

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Content (translated into English)

Machine translation of [usa-25-barnett-lost-translation-exploiting-unicode-compressed.md](#), which holds the source's own words. Code, payloads, type names, URLs and CVE identifiers were masked before translating and restored after, so they are byte-identical to the original.

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Lost In : Exploiting Unicode Normalization Ryan Barnett Isabella Barnett

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Telephone Game
Bad patch is Backpack is gonna be at Blackhat is gonna be off noon!? gonna be soon..? awesome!

#BHUSA @BlackHatEvents "Web" Telephone Game

CDN Content Delivery Network

#BHUSA @BlackHatEvents CDN Content Delivery Network

#BHUSA @BlackHatEvents CDN Content Delivery Network

#BHUSA @BlackHatEvents CDN Content Delivery Network

#BHUSA @BlackHatEvents CDN Content Delivery Network

#BHUSA @BlackHatEvents CDN Content Delivery Network

#BHUSA @BlackHatEvents CWE-180: Incorrect Behavior Order: Validate Before Canonicalize

CDN Content Delivery Network

#BHUSA @BlackHatEvents Agenda Decoding Errors Truncation Confusables Casing Combining
Diacritics #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Internationalization (i18n) ABC 1 2

ç é ã â #BHUSA @BlackHatEvents <https://sonarsource.github.io/utf8-visualizer/>

Unicode Codepoint

#BHUSA @BlackHatEvents <https://sonarsource.github.io/utf8-visualizer/>

Unicode Codepoint

Raw Bytes

#BHUSA @BlackHatEvents <https://sonarsource.github.io/utf8-visualizer/>

Unicode Codepoint

Raw Bytes

UTF-8 Encoding

#BHUSA @BlackHatEvents Unicode ASCII American Standard Code for Universal Character Encoding Information Exchange

Wide range of characters 128 Characters

8, 16 or 32 bits 7 bits

Requires more storage space Requires less storage space

#BHUSA @BlackHatEvents Agenda Decoding Errors Multi-Byte Aware?

CWE-172: Encoding Error

#BHUSA @BlackHatEvents Leading bits signal character byte length

#BHUSA @BlackHatEvents "Mojibake" Output

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Not Multi-byte aware = Mojibake

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Multi-byte aware = Correct Decoding

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Taint Tracking

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Agenda Decoding Errors Multi-Byte Aware?

Overlong Encoding

CAPEC-80: Using UTF-8 Encoding to Bypass Validation Logic #BHUSA @BlackHatEvents Needless Packaging

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents 7654321

#BHUSA @BlackHatEvents Padding

7654321 7 654321

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Overlong spaces

Overlong < #BHUSA @BlackHatEvents Cannot handle overlong

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Agenda Decoding Errors Truncation

#BHUSA @BlackHatEvents Bug Bounty CRLF Injection

<https://thecyberneh.github.io/posts/MicrosoftBugbounty/> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents

<https://gchq.github.io/CyberChef/> #BHUSA @BlackHatEvents MSB LSB Byte 2 Byte 1 Byte 0 Byte 2 Byte 0 Byte 1 Byte 1 Byte 0 Byte Ordering Byte 2 Little Endian Big Endian #BHUSA @BlackHatEvents Single Byte Buffer

#BHUSA @BlackHatEvents Single Byte Buffer

Byte 2 Byte 1 Byte 0

#BHUSA @BlackHatEvents Byte 2 Byte 1 Byte 0 Byte 0

#BHUSA @BlackHatEvents Byte 2 Byte 1 Byte 0 Byte 0

#BHUSA @BlackHatEvents <https://gchq.github.io/CyberChef/> #BHUSA @BlackHatEvents Byte Truncation

<https://gchq.github.io/CyberChef/> #BHUSA @BlackHatEvents <https://shazzer.co.uk/unicode-table> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Agenda Decoding Errors Truncation Confusables

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents I don't accept Greek

#BHUSA @BlackHatEvents But that looks a lot like this...

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents So I'm just gonna use it instead

#BHUSA @BlackHatEvents That's not gonna cause any problems

#BHUSA @BlackHatEvents Right...?

#BHUSA @BlackHatEvents ...

#BHUSA @BlackHatEvents Form Description Example

NFD Canonical Decomposition

Canonical Decomposition NFC followed by Canonical Composition

NFKD Compatibility Decomposition

Compatibility Decomposition, NFKC followed by Canonical Composition #BHUSA @BlackHatEvents

<https://util.unicode.org/UnicodeJsps/idna.jsp> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents

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@BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents

<https://unicode.org/Public/MAPPINGS/VENDORS/MICSFT/WindowsBestFit/bestfit1251.txt> #BHUSA @BlackHatEvents =

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents <https://worts.fit/> #BHUSA @BlackHatEvents

<https://util.unicode.org/UnicodeJsps/confusables.jsp> #BHUSA @BlackHatEvents #BHUSA

@BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents

#BHUSA @BlackHatEvents Agenda Decoding Errors Truncation Confusables Casing

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA

@BlackHatEvents #BHUSA @BlackHatEvents <http://www.unicode.org/charts/case/> #BHUSA

@BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents

#BHUSA @BlackHatEvents Agenda Decoding Errors Truncation Confusables Casing Combining Diacritics #BHUSA @BlackHatEvents NFD

#BHUSA @BlackHatEvents <https://blog.voorivex.team/puny-code-0-click-account-takeover> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Collation Algorithm Version

#BHUSA @BlackHatEvents Accent Insensitive & Case Insensitive

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents True True

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents False

#BHUSA @BlackHatEvents <https://www.unicode.org/charts/collation/index.html> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Unicode Collation Algorithm

= #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents <https://ozb2apmi.xssy.uk/> #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Container Element

#BHUSA @BlackHatEvents Server Side FreeMarker Template Language

Concatenate input with tag and then normalize

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents =

#BHUSA @BlackHatEvents =

#BHUSA @BlackHatEvents #BHUSA @BlackHatEvents #BHUSA @BlackHatEvents Key Takeaways

1 Normalize before security logic

2 Validate URL decodings

3 Confirm normalization forms in use

#BHUSA @BlackHatEvents